

PROJECT REPORT
ON
“DAIRY MANAGEMENT SYSTEM”

Submitted for the partial fulfillment of the Requirement for the award of degree
Bachelor of Computer Applications



Uttaranchal School of Computing Sciences
USCS
(Session : 2025-2026)

Under the provision of

Mr. Deepak kumar
Assistant professor
USCS

Submitted by:

Nitin kumar
BCA II 'C'
UU2509000246

ACKNOWLEDGEMENT

The most awaited moment of any endeavor is successful completion but nothing can be done successfully if done alone. Success is the outcome of contribution and consistent help of various persons and we thank those ones who helped us in successful completion of this project.

Primarily I would like to thank **Prof (Dr.) Sonal Sharma, Director-USCS** for providing a healthy and encouraging environment to study.

I profusely thankful to **Dr. Sameer Dev Sharma HOD-USCS** for providing all the required resources in the successful completion of my project report.

I express my thanks to **Mr. Deepak kumar, Project Mentor, Assistant Professor-USCS**. She has been generous enough to provide me an opportunity and accepting my candidature for the most valuable guidance and affordable treatment given to us at every stage to boost my moral.

Nitin kumar
Roll no. - 51
B.C.A 2nd Sem

DECLARATION

I hereby declare that the project report entitled **Dairy Management System** submitted by **Nitin kumar** to Uttaranchal School of Computing Sciences. The project report was done under the guidance of **Mr. Deepak kumar, Assistant Professor-USCS**. I further declare that the work reported in this project report has not been submitted and will not be submitted, either in part or in full, for the award of any other degree or diploma in this university or any other university or institute.

Nitin kumar
Roll no.-51
B.C.A 2nd Sem

CERTIFICATE OF ORIGINALITY

This is to certify **Nitin Kumar** student of BCA 2nd semester, of **Uttaranchal School of Computing Sciences, Dehradun**, has completed the Project Report for the topic **Dairy Management System** for the session (2025 – 2026).

Under the guidance of:
Mr. Deepak kumar
Assistant Professor
-USCS

TABLE OF CONTENT

S.No.	Contact Name	Page No.
1.	INTRODUCTION	6
2.	SCOPE OF PROJECTS	7
3.	PROFILES OF TWO IT COMPANIES	
4.	REPORT OF PROJECT	
5.	PROJECT PLANNING	
6.	SOFTWARE REQUIREMENTS SPECIFICATION	
7.	HARDWARE REQUIREMENTS	
8.	FLOW CHART	
9.	STATE DIAGRAM	
10.	E-R DIAGRAM	
11.	SYSTEM DESIGN	
12.	FEASIBILITY STUDY	
13.	SYSTEM ANALYSIS	
14.	CODING	
15.	OUTPUT	
16.	TESTING	
17.	FUTURE SCOPE	
18.	BIBLIOGRAPHY	

VV

INTRODUCTION

We are creating a Window application for “**Dairy management system**”.

In C++ language by using Code block & Dev C++. The mission of the Milk Production House Project is to create to communication between rural area people and dairy management. Our goal is to develop this application to encourage our dairy industry. Dairy milk management system is a software application to maintain day to day transactions in a Milk dairy. This software helps to register all the suppliers, Buyer details, purchase, Sales details. Our plan is to collect the Milk from many different sources and distribute the same to the Milk buyers.

SCOPE OF PROJECTS

Dairy management system is user friendly software.

Reduces the time and work efforts by humans and to automate the dairy.

User can see the previous transactions and view milk rates.

Easy to maintain and store records and to generate bills.

Records or the data of a user can be easily fetched.

PROFILES OF TWO IT COMPANIES

Tata Consultancy Services (TCS)

Tata Consultancy Services (TCS) is one of the world's largest and most trusted Information Technology (IT) services, consulting, and business solutions companies. Established in **1968**, TCS is a flagship company of the **Tata Group**, India's most respected business conglomerate. Its headquarters is located in **Mumbai**, and the company operates across more than **150 locations in 46+ countries**, serving global clients across diverse industries such as banking, financial services, insurance, healthcare, retail, telecom, manufacturing, energy, and government sectors. Over the years, TCS has built a strong reputation for delivering high-quality IT solutions, business transformation services, and advanced digital technologies. The company is valued worldwide for its research-driven approach, mainly through **TCS Innovation Labs**, which work on cutting-edge technologies and patents. This innovation culture has helped TCS consistently rank among the world's most valuable IT services brands.

TCS offers a broad portfolio of services and solutions that cater to global business needs. Its **IT Services & Consulting** division provides end-to-end strategic and technical support. In **Application Development and Maintenance**, TCS designs, builds, and manages large-scale enterprise applications that support business operations. It is also a major provider of **Cloud Computing services**, helping organizations migrate to leading cloud platforms such as AWS, Microsoft Azure, and Google Cloud. TCS delivers AI-powered solutions, including predictive analytics, automation tools, and chatbots, enabling companies to adopt intelligent systems. In the field of **Cybersecurity**, TCS provides essential services like threat detection, identity management, and risk assessment to protect organizations from cyber threats. The company also specializes in **Enterprise Resource Planning (ERP)** solutions such as SAP and Oracle, helping industries streamline operations. Additionally, TCS runs extensive **Business Process Outsourcing (BPO)** services such as customer support and financial processing, enabling companies to reduce costs and improve operational efficiency.

The company has deep expertise in modern and emerging technologies. It works with a wide set of **programming languages**, including Java, Python, C++, and C#. Its teams are skilled in advanced **frameworks and platforms** like Spring Boot, .NET, React, Angular, and Node.js. TCS is also a major player in cloud technologies, with certified experts on AWS, Azure, and Google Cloud. Its database capabilities include SQL-based and NoSQL systems such as Oracle,

MySQL, and MongoDB. Beyond traditional technologies, TCS invests heavily in advanced fields like **Blockchain**, **IoT (Internet of Things)**, **Robotic Process Automation (RPA)**, and **Big Data Analytics** using tools such as Hadoop and Spark. The company's internal learning platforms ensure that employees continuously upgrade their skills to keep pace with evolving industry demands.

TCS is equally known for its strong and employee-friendly **work culture**. It offers excellent training programs for fresh graduates, including TCS Xplore, which prepares newcomers for realworld projects. Employees enjoy a healthy work-life balance and opportunities to work on international projects in countries like the USA, UK, Europe, and Australia. The company encourages teamwork, creativity, innovation, and continuous learning. With various internal certification programs and leadership development tracks, TCS provides ample opportunities for career growth.

The company's **placement process** is structured and transparent. For eligibility, students must have a minimum of **60% marks** throughout their academic history and must not have any active backlogs. The selection process usually includes four rounds: an online aptitude and logical reasoning test, a technical interview focusing on programming and problem-solving skills, a managerial interview in some cases, and finally an HR interview to assess communication skills and personality. This multi-step process ensures that the company recruits candidates with strong analytical ability, technical knowledge, and professional behavior.

For communication, TCS can be contacted through email at **corporate.tcs@tcs.com**, phone at **+91-22-6778 9999**, or via its official LinkedIn page.

Infosys Limited

Infosys Limited is a leading global information technology, consulting, and outsourcing company founded in **1981** and headquartered in **Bengaluru, India**. Operating in more than **50 countries**, Infosys helps organizations transform into digital and data-driven enterprises. The company is widely known for its strong focus on innovation, sustainability, and advanced technology research.

Its **Global Education Center in Mysore** is one of the world's largest corporate training campuses, providing high-quality technical and professional training to freshers. Initiatives like **Infosys Labs** and the AI platform **Infosys Nia** strengthen its position in digital intelligence and automation. Infosys offers a diverse range of services, including **consulting, software development, maintenance, cloud and infrastructure services, and cybersecurity**. The company specializes in digital transformation through **AI, ML, automation, and UI/UX design**. Additionally, Infosys implements enterprise systems such as **SAP, Oracle, and CRM platforms**, helping businesses improve efficiency and adopt modern technologies.

The company works with a wide range of technologies, including programming languages like **Python, Java, JavaScript, Kotlin, and Swift**, and front-end frameworks such as **Angular, React, and Vue**. It has strong expertise in cloud platforms like **AWS, Azure, and GCP**, and uses data analytics tools such as **Power BI, Tableau, and Hadoop**. Infosys also excels in automation tools like **UiPath** and **BluePrism**, and advanced AI frameworks like **TensorFlow** and **PyTorch**. Employees continuously enhance their skills through the internal learning platform **Lex**.

Infosys is known for its disciplined, structured, and employee-friendly work culture. Freshers receive mentorship and opportunities for onsite projects abroad. The company follows a transparent placement process with an online assessment, technical interview, and HR interview. Candidates need 60% in academics with no active backlogs.

For communication, Infosys can be reached via email at **askus@infosys.com**, by phone at **+9180-2852 0261**, or through their official LinkedIn page.

REPORTS OF PROJECT

EXISTING SYSTEM

When we Analysis the Manage about this firm then we face that they working with manual. And we all know that the manual system has many disadvantages. Some are mentioned below...

- The manual system requires more time for processing.
- It requires more critical work.
- The manual system is more error prone.
- Difficult to maintain.
- Manual system is costly.
- Immediate response to the queries is difficult and time consuming.
- More men power needed.
- Manual system show of the particular place.

PROPOSED SYSTEM

- No Multiuser environment is provided. In this when one user is updating a form; the other user cannot open the same form to avoid conflicts.
- Security to the data is provided by means of Login Form. Only authorized users can have access to the system.
- The system allows users to maintain records of customers, suppliers, their orders, raw materials, stock availability as well as bill generation.
- This system also allows users to generate Customer reports, Supplier reports, sales and purchase reports as well as raw material's reports in the form of crystal reports.
- Easy to search and update Customer or supplier records by their Id.

SOFTWARE REQUIREMENTS SPECIFICATION

A **Dairy Management System** built using C++ requires specific software and hardware components to ensure smooth operation. The software components provide the necessary tools for coding, compiling, and database management, while the hardware components ensure efficient performance in a real-world dairy store environment.

Software Requirements

1. Operating System (OS)

The Grocery Management System can run on multiple operating systems, including:

- **Windows 10/11** – Commonly used for development and deployment.
- **Linux (Ubuntu, Fedora, etc.)** – Preferred for server-based applications.
- **macOS** – Used by developers who prefer Mac for C++ programming.

2. Programming Language

- **C++ (Standard C++17 or later)** is the core language used for developing the system.
- It provides high performance and efficient memory management, making it suitable for real-time applications.

3. Integrated Development Environment (IDE)

To write, compile, and debug C++ code, the following IDEs can be used:

- **Code::Blocks** – Lightweight and easy to use.
- **Dev-C++** – Suitable for beginners.
- **Microsoft Visual Studio** – Best for large-scale projects with GUI.
- **Qt Creator** – Ideal for GUI-based applications using the Qt framework.

4.Compiler

GCC (GNU Compiler Collection) – Used in Linux/macOS environments.

MinGW (Minimalist GNU for Windows) – Commonly used for Windows development. **MSVC**

(Microsoft Visual C++ Compiler) – Comes with Visual Studio.

5.Database Management System (DBMS)

A database is required to store product details, customer data, and sales records. Some commonly used databases are:

MySQL – Recommended for large-scale data handling.

SQLite – Lightweight, file-based database suitable for small businesses. **PostgreSQL** – Advanced database with strong security features.

Libraries and Frameworks

Qt Framework – Used for building a graphical user interface (GUI).

MySQL Connector for C++ – Allows integration with MySQL databases.

Boost Library – Provides additional C++ functionalities.

OpenCV (Optional) – Can be used for barcode scanning or facial recognition.

Additional Software

XAMPP – A server environment for running MySQL databases locally.

Git – Used for version control and collaborative development.

HARDWARE REQUIREMENTS

Minimum Hardware Requirements

For basic functionality, the system can run on standard hardware with the following specifications:

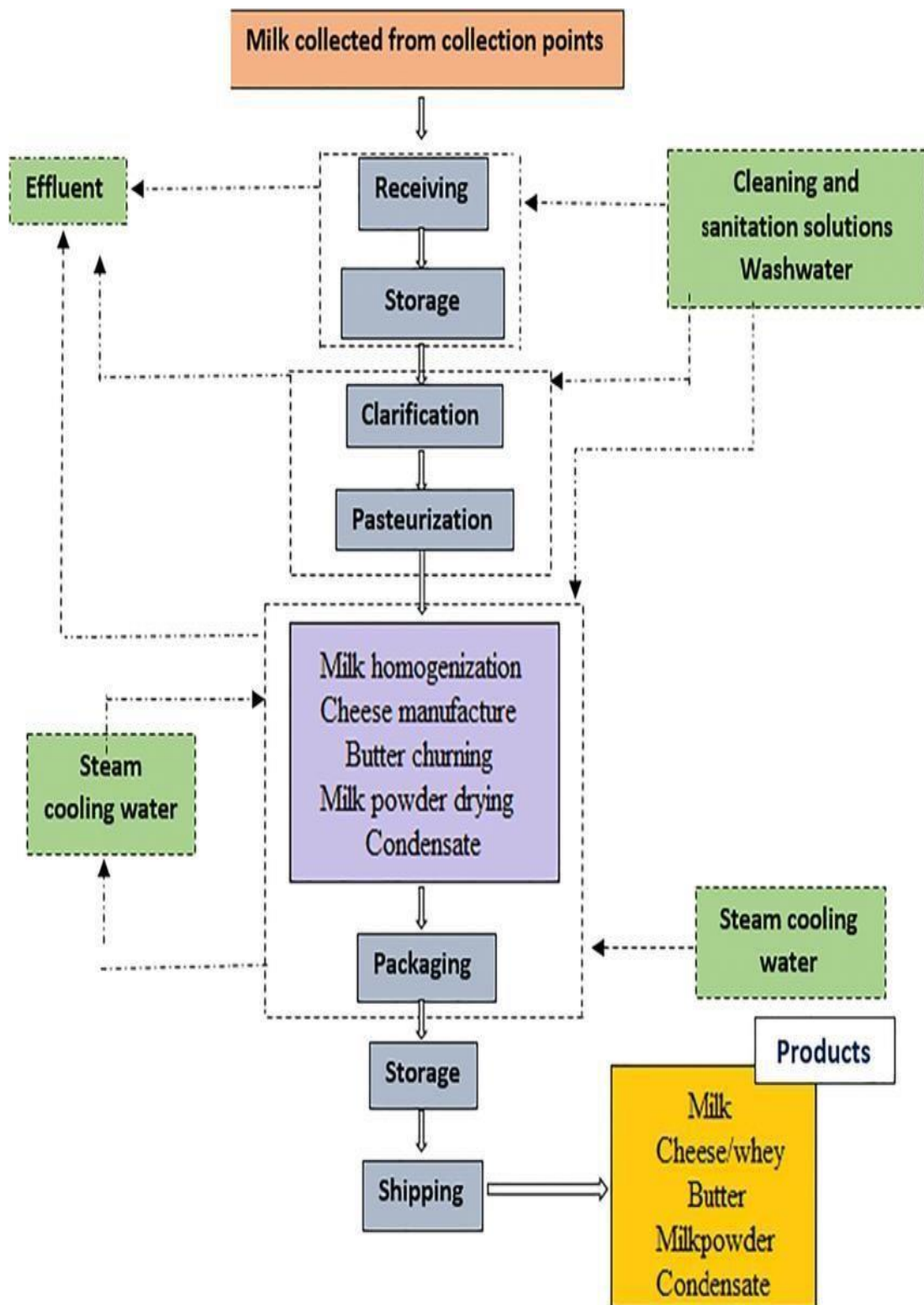
Component	Specification
Processor (CPU)	Intel Core i3 / AMD Ryzen 3
RAM	4GB (Minimum) / 8GB (Recommended)
Storage	256GB HDD / 128GB SSD
Monitor	1366x768 resolution (Minimum)
Keyboard & Mouse	Standard input devices

Recommended Hardware for Large Grocery Stores

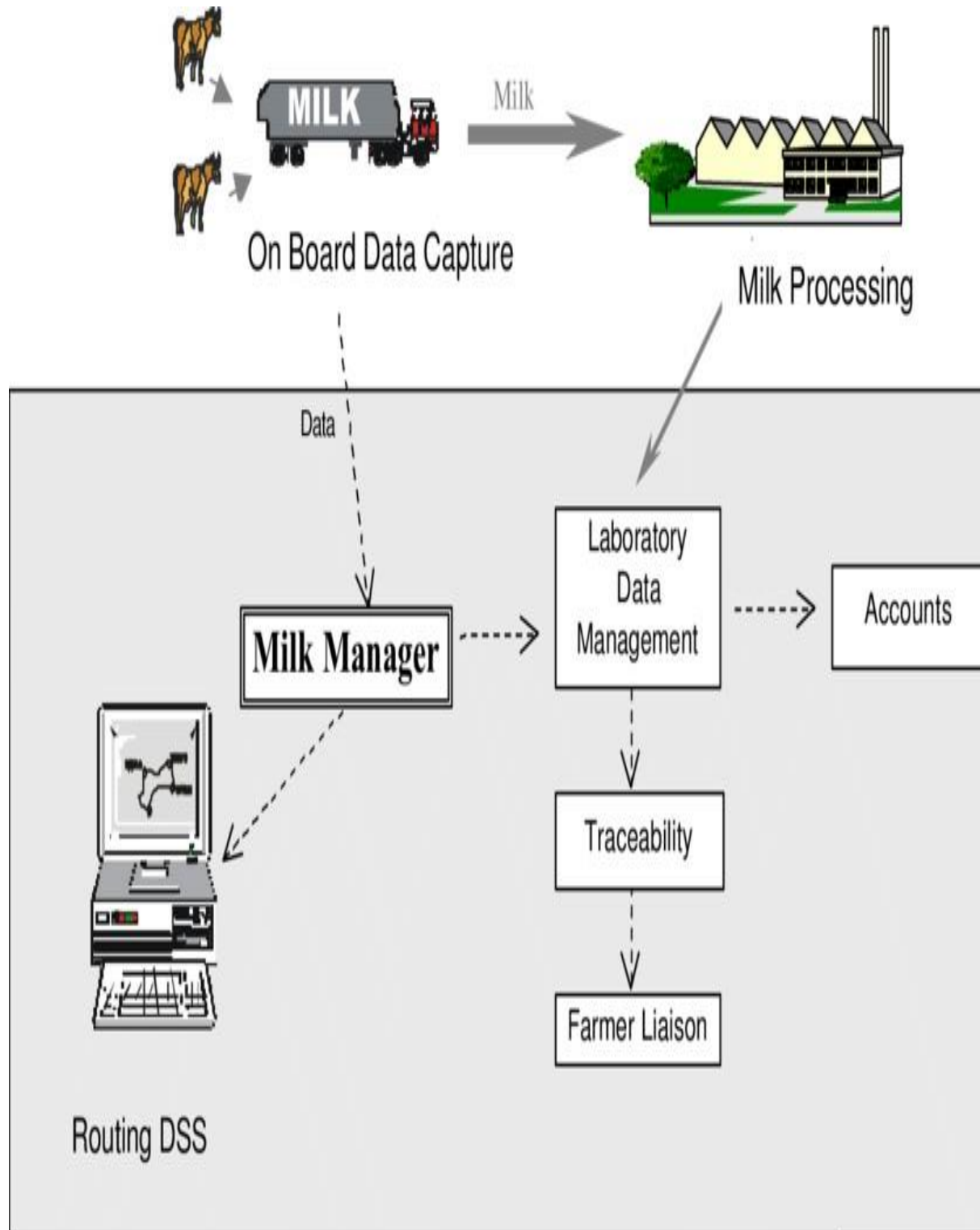
For businesses with high transaction volumes, a more powerful setup is needed

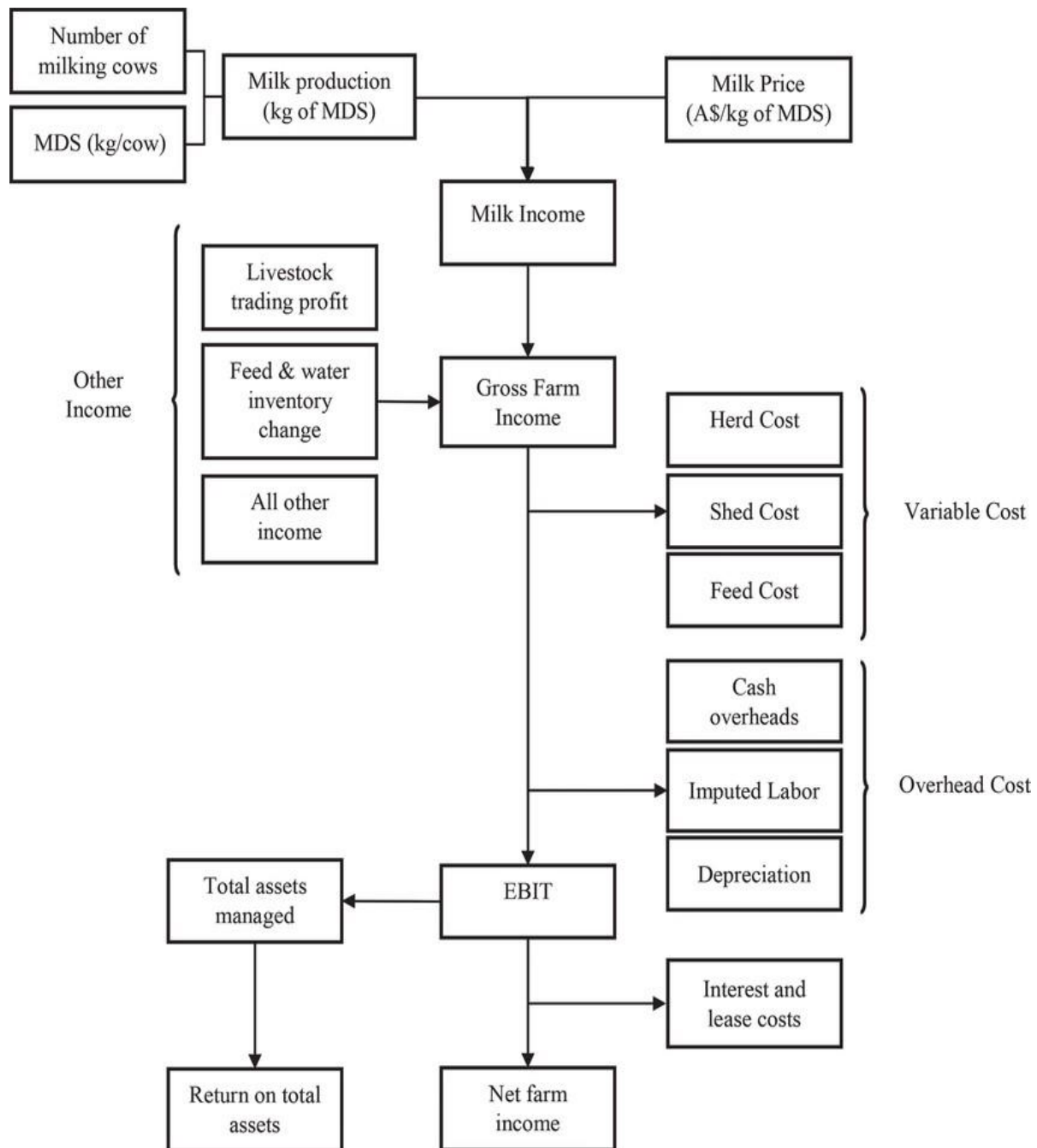
Component	Specification
Processor (CPU)	Intel Core i5/i7 or AMD Ryzen 5/7
RAM	8GB – 16GB
Storage	512GB SSD (For better speed)
Multi-Monitor Setup	Dual monitors (For cashier and admin use)
Touchscreen POS System	For easy billing and checkout
Barcode Scanner	USB/Handheld scanner for fast product entry
Thermal Receipt Printer	For generating customer bills
UPS (Uninterruptible Power Supply)	To prevent data loss during power failures

FLOWCHART OF DMS

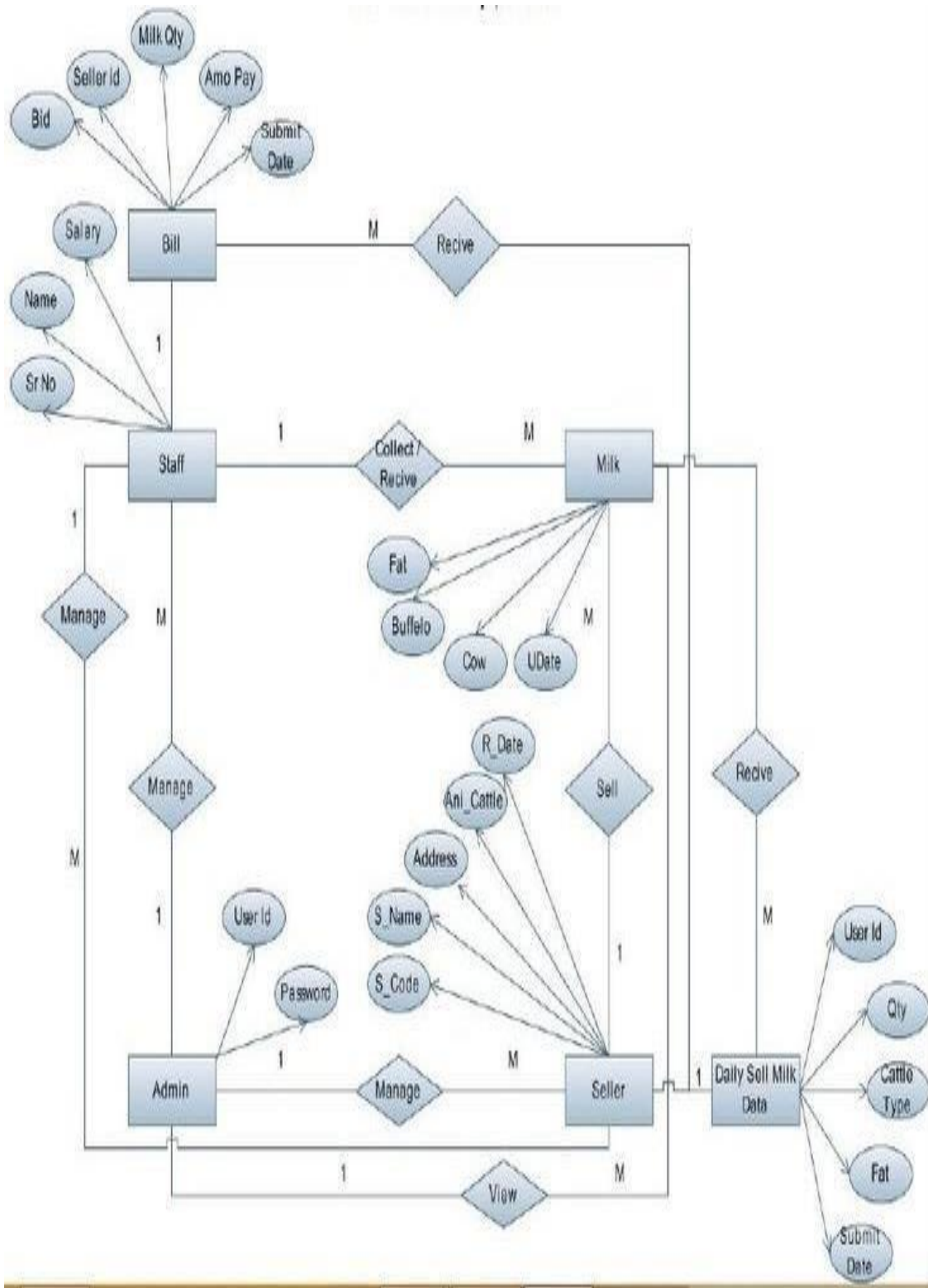


STATE DIAGRAM OF DMS





E-R DIAGRAM OF DMS



SYSTEM DESIGN OF DMS

ADMIN

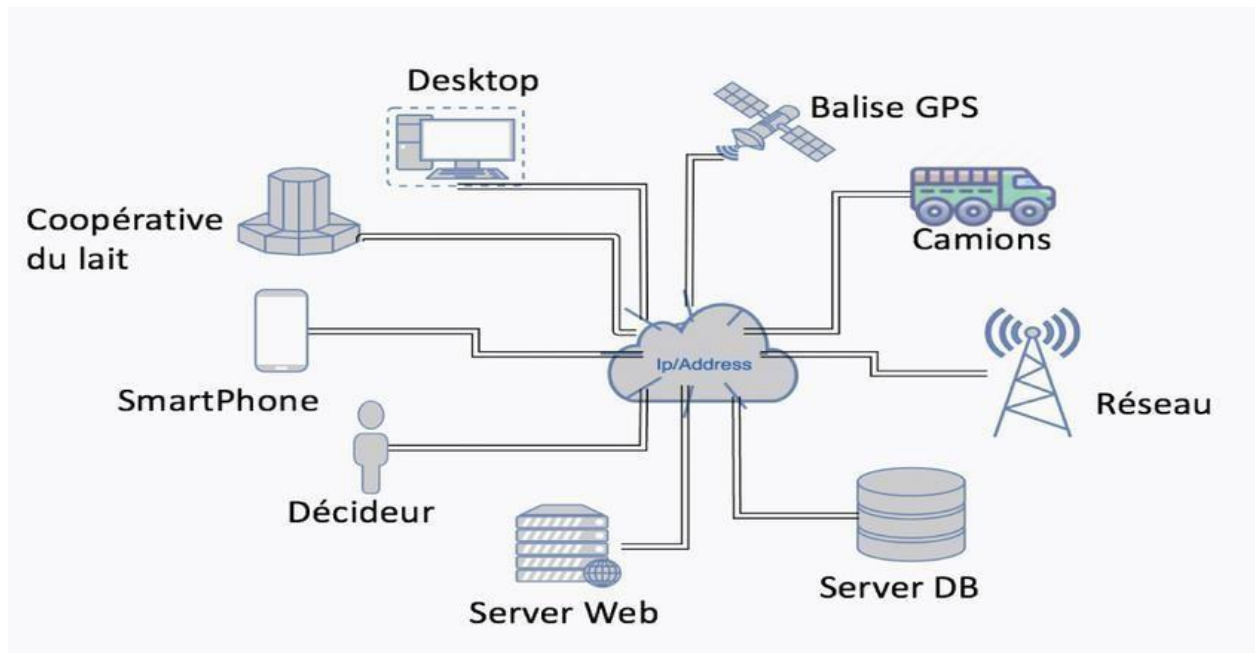
- ☐ Login
- ☐ Manages rates
- ☐ Manages staff
- ☐ Manages seller

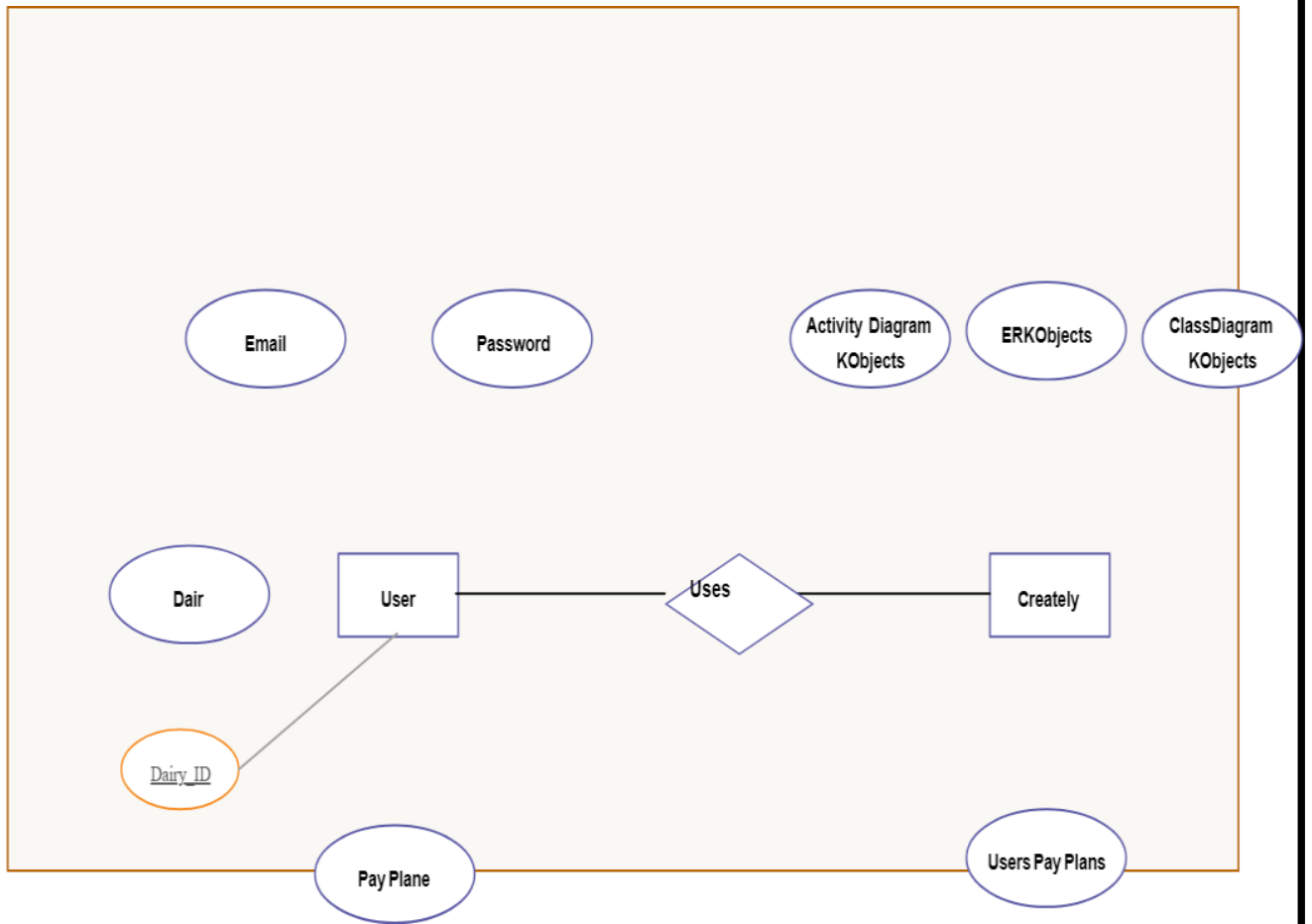
STAFF

- ☐ Login
- ☐ Seller Registration
- ☐ Manage feedback
- ☐ Insert milk details
- ☐ Seller
- ☐ Edit products
- ☐ Exit

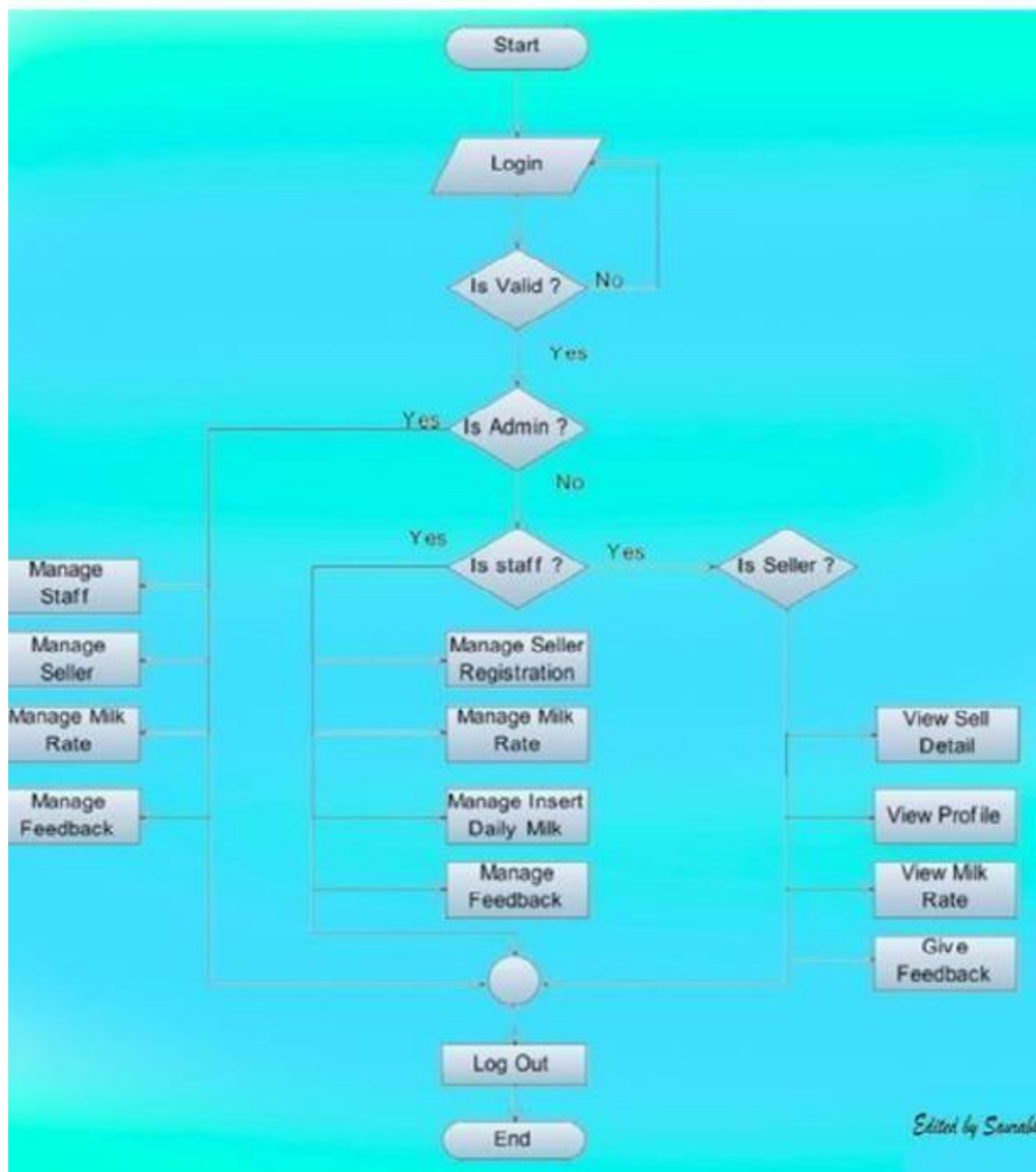
USER

- ☐ Registration
- ☐ Login
- ☐ View rates
- ☐ Feedback





SYSTEMATIC FLOWCHART OF DMS



FEASIBILITY STUDY

A **feasibility study** evaluates whether the **Dairy Management System** is practical, costeffective, and beneficial for implementation. It analyzes **technical, operational, economic, legal, and schedule feasibility** to determine the system's viability.

1. Technical Feasibility

Programming Language – The system is developed in C++, which is efficient, fast, and widely used.

File Handling – Uses **text files** to store inventory, billing, customer, and supplier data, eliminating the need for complex databases.

Hardware Requirements – Runs on any standard computer with basic specifications (Processor: Intel i3+, RAM: 2GB+).

Software Requirements – Requires a C++ **compiler** (e.g., GCC, Turbo C++, or Microsoft Visual Studio).

2. Operational Feasibility

User-Friendly Interface – The system features a **simple menu- based interface**, making it easy for store employees to operate.

Efficient Inventory Management – Reduces manual work and human errors in stock tracking.

Fast & Accurate Billing – Ensures **quick calculations** and prevents overcharging or undercharging customers.

Data Storage & Retrieval – Stores customer and supplier records for easy future reference.

3. Economic Feasibility

- Low Development Cost – Built using C++, a free and open- source language, reducing licensing costs.
- No Need for Expensive Databases – Uses text files for data storage instead of paid database solutions.
- Minimal Hardware Investment – Runs on existing computers, avoiding additional hardware costs.
- Improves Business Efficiency – Saves time and labor costs by automating inventory and billing.

4. Schedule Feasibility

- Development Timeline – The project can be developed and implemented within 3–4 weeks.
- Easy Maintenance – Since it uses structured code and file handling, updates and bug fixes can be done quickly. Future Enhancements

5. Legal Feasibility

- **Compliance with Software Licensing** – Uses C++, an open- source programming language, ensuring no copyright issues.
- **Data Privacy & Security** – The system stores **basic customer and supplier details** without violating privacy laws.
- **Legally Acceptable Billing System** – Provides **accurate calculations and recordkeeping**, preventing fraud or tax-related issues.

6. Feasibility Study Conclusion

- The **Dairy Management System in C++** is **highly feasible** in all aspects:
- **Technically Feasible** – Runs on basic hardware and software.
- **Operationally Feasible** – Automates store operations, reducing errors and improving efficiency.
- **Economically Feasible** – Low-cost development with high business benefits.
- **Schedule Feasible** – Can be developed in a short period with room for future enhancements.
- **Legally Feasible** – Complies with licensing and privacy regulations.

SYSTEM ANALYSIS

Introduction

The **Dairy Management System** is designed to streamline the operations of a dairy store by automating inventory management, billing, and customer-supplier record keeping. It reduces manual errors, improves efficiency, and enhances business operations. This system analysis explores its **requirements, feasibility, components, and operational flow**.

Problem Statement

Managing a dairy store manually can lead to several challenges, including:

Inaccurate Inventory Tracking – Difficulty in keeping track of stock levels.

Billing Errors – Manual calculation mistakes can lead to financial losses.

Data Management Issues – Maintaining customer and supplier records manually is inefficient.

Time-Consuming Processes – Manual bookkeeping slows down operations.

System Objectives

Automate Inventory Management – Add, remove, and update items efficiently.

Generate Bills Instantly – Calculate total costs based on item price and quantity

Store Customer & Supplier Details – Maintain records for future reference.

Generate Sales Reports – Provide insights for business decision-making.

Feasibility Study

Technical Feasibility – The system is developed using **C++**, a widely used programming language, and uses **file handling for data storage**, making it suitable for small to medium businesses.

Operational Feasibility – The system is **user-friendly** with a simple menu-based interface that does not require extensive training.

Economic Feasibility – Since it is built using **open-source tools (C++, text files)**, there is **no extra cost** for implementation.

Schedule Feasibility – The system can be developed in **a few weeks**, making it a **quick solution** for store management.

System Components

User Interface – A command-line interface for users to interact with the system.

File Handling System – Used for storing inventory, customer, and supplier data.

Inventory Module – Allows adding, viewing, and updating grocery items.

Billing Module – Calculates bills and updates stock after purchases.

Report Generation Module – Generates reports for sales and inventory.

Customer & Supplier Management – Stores and retrieves customer/supplier details.

System Flowchart

1. User selects an option from the **menu**. 2. Based on selection:
Inventory is updated (Add/Delete/View items).

Billing is processed (Checks stock, calculates total, updates inventory).

Customer/Supplier details are stored/retrieved.

Sales reports are generated saves data to files for future use.

7. Advantages of the System

Eliminates manual errors in inventory and billing.

Fast and efficient – Saves time in store operations.

Easy to use – Simple interface requiring minimal training.

Data is securely stored in files for future reference.

8. Limitations & Future Enhancements

No GUI – Can be improved using **Qt or Tkinter**.

No database – Can integrate **MySQL or SQLite** for better data management.

No multi-user access – Can be expanded for **cloud-based use**.

No real-time tracking – Can integrate **barcode scanning and AI- based forecasting**.

CODING

```
#include <iostream> #include
    <vector>
#include <string>

using namespace std;

class Cow { public:
    string
    name;   int
    age;
    double milkProduction; // in liters

    Cow(string n, int a, double mp) : name(n), age(a), milkProduction(mp) {}
};

class DairyFarm { private:
    vector<Cow> cows;

public:
    void addCow(const      string&      name, int      age,      double
        milkProduction) cows.emplace_back(name, age, milkProduction); cout << "Cow
        " << name << " added successfully!" << endl;
    }
```

```

void displayCows() { if
    (cows.empty()) { cout << "No cows in the farm."
        << endl; return;
    }
    cout << "List of Cows:" << endl; for
    (const auto& cow : cows) { cout << "Name: " << cow.name << ", Age: " << cow.age << ",
        Milk Production: " <<
cow.milkProduction << " liters" << endl; }
    }

void totalMilkProduction() { double total
    = 0; for (const auto& cow : cows) {
        total += cow.milkProduction;

    } cout << "Total Milk Production: " << total << " liters" <<
endl; }

};

void displayMenu() { cout << "\nDairy Management System"
    << endl; cout <<
    "1. Add Cow" << endl;
    cout << "2. Display Cows" << endl;
    cout << "3. Total Milk Production" << endl; cout <<
    "4. Exit" << endl; cout <<
    "Choose an option: ";

```

```
}
```

```
int main() {
```

```
    DairyFarm farm;
```

```
    int choice;
```

```
    do {
```

```
        displayMenu();
```

```
        cin >> choice;
```

```
        switch (choice) {
```

```
            case 1: { string
```

```
                name; int
```

```
                age; double
```

```
                milkProduction;
```

```
                cout << "Enter Cow Name: "; cin
```

```
                >> name; cout << "Enter Cow
```

```
                Age: "; cin
```

```
                >> age;
```

```
                cout << "Enter Milk Production (in liters): "; cin >>
```

```
                milkProduction;
```

```
            farm.addCow(name, age, milkProduction); break;
```

```
        }
```

```
    case 2:
```

```
        farm.displayCows(); break;
    case 3:
        m.totalMilkProduction(); break;
    case 4:
        farcout << "Exiting the system." << endl; break;
    default:
        cout << "Invalid choice. Please try again." << endl;
    }
} while (choice != 4);

return 0;
}
```

OUTPUT

```
1 Dairy Management System
2 1. Add Cow
3 2. Display Cows
4 3. Total Milk Production
5 4. Exit
6 Choose an option: 1
7 Enter Cow Name: Bessie
8 Enter Cow Age: 5
9 Enter Milk Production (in liters): 20
10 Cow Bessie added successfully!
11
12 Dairy Management System
13 1. Add Cow
14 2. Display Cows
15 3. Total Milk Production
16 4. Exit
```

```
17 Choose an option: 1
18 Enter Cow Name: Daisy
19 Enter Cow Age: 4
20 Enter Milk Production (in liters): 15
21 Cow Daisy added successfully!
22
```

```
37 4. Exit
38 Choose an option: 3
39 Total Milk Production: 35 liters
40
41 Dairy Management System
42 1. Add Cow
43 2. Display Cows
44 3. Total Milk Production
45 4. Exit
```

```
22
23 Dairy Management System
24 1. Add Cow
25 2. Display Cows
26 3. Total Milk Production
27 4. Exit
28 Choose an option: 2
29 List of Cows:
30 Name: Bessie, Age: 5, Milk Production: 20 liters
31 Name: Daisy, Age: 4, Milk Production: 15 liters
32
33 Dairy Management System
34 1. Add Cow
35 2. Display Cows
36 3. Total Milk Production
37 4. Exit
```


TESTING

1. FUNCTIONAL TESTING

Focuses on checking individual features to ensure they work as intended

Test Case ID	Feature	Test Description	Expected Outcome	Status
TC001	Add Item	Add a valid item to the inventory.	Item appears in inventory.txt with correct details.	
TC002	Add Item	Add an item with a negative price or quantity.	System rejects invalid input.	
TC003	Display Items	Display inventory with multiple items.	All items appear with proper formatting.	
TC004	Generate Bill	Purchase valid item with available stock.	Bill generated with correct total. Stock reduces accordingly.	
TC005	Generate Bill	Attempt to purchase an item that doesn't exist.	Error message: "Item not found!"	

TC006	Generate Bill	Attempt to purchase an item with insufficient stock.	Error message: "Not enough stock!"
TC007	Add Customer	Add a customer with a valid phone number.	Customer saved in customers.txt.
TC008	Add Customer	Enter an invalid phone number (less than 10 digits).	Error message: "Invalid phone number!"
TC009	Add Supplier	Add a valid supplier with a contact number.	Supplier appears in suppliers.txt.
TC010	Exit System	Select exit option.	Program terminates gracefully.

FUTURE SCOPE OF THE DMS

As now the project/software will work upon command property in future we will make GUI (graphical User Interface) or web-based software/system so that every user or employ can access it remotely.

- Seller can get his/her payment directly into account.
- Customer also be a user of system.
- Performance will be improved day to day users increment.
- In future we can convert it in android application (APP).
- Seller can see purchase sell milk show to online.
- Seller can her payment direct in her account.

BIBLIOGRAPHY

WEBSITES:

1. www.wikipedia.org
2. www.codeguru.com
3. www.stackoverflow.com
4. www.youtube.com
5. www.microsoft.com
6. www.codeproject.com

Reference:

Acharya, Kamal. "STUDENT INFORMATION MANAGEMENT SYSTEM." Authorea Preprints (2023).

Acharya, Kamal. "Library Management System." Available at SSRN4807104 (2019). ACHARYA, KAMAL, et al. "LIBRARY MANAGEMENT SYSTEM." (2019).

Acharya, Kamal. "Online bus reservation system project report." Authorea Preprints (2024). Acharya, Kamal. "Online bus reservation system project report." (2024).

Acharya, Kamal. "Online Bus Reservation System." SSRN ElectroNIC ASIA Journal (2024): n. pag. Acharya, Kamal. "Student Information Management System Project."

SSRN ElectroNIC ASIA Journal (2024): n. pag. Acharya, Kamal. "ATTENDANCE MANAGEMENT SYSTEM." International Research Journal of Modernization in Engineering Technology and Science (2023):

n. pag. Acharya, Kamal. "College Information Management System." SSRN ElectroNIC ASIA Journal (2024): n. pag. Acharya, Kamal, Attendance Management System Project (April 28, 2024).

- No delay in calculating WPM ○

Database saves data instantly

2. Exception/Error Handling Testing

- Tested how the system handles invalid conditions □

Example:

- Empty text submission ○ Non-alphabet input ○ App closing before completing

Testing Strategy Used

1. Requirements-Based Testing Strategy

- All tests were designed based on project requirements:
- Typing measurement
- Timing system
- Accuracy calculation
- Result saving

2. Module-Wise Testing Strategy

- Each module was tested separately: ○ Paragraph display module ○ Paragraph loads correctly ○ No changes allowed by user

Timer Module

- Timer starts only when Start button is pressed □ Timer stops when Submit is clicked

Typing Input Module

- Records user input properly
- Handles backspace and mistakes

Result Calculation Module

- Correct WPM formula
- Accuracy calculation working properly

Database/Storage Module

- Results correctly saved
- No duplicate saved

3. Incremental Testing Strategy

- Testing was performed step-by-step while developing features.
- Avoids big errors at the end of development

4. User Acceptance Testing (UAT)

- Real user tested the typing application □ Ensured:
 - App is easy to use
 - Results are understandable
 - UI is comfortable

5. Regression Testing

- After adding new features, previously working features were re-tested.
- Ensures new updates do not break old functionality.

Testing Reports

1. Unit Testing

Unit testing was carried out to verify individual components of the system. Each function and class was tested independently to ensure that it worked as expected.

Modules Tested:

- **Typing Logic Module:** Tested for accurate calculation of words per minute (WPM) and accuracy percentage.
- **Timer Module:** Verified that the timer starts and stops correctly and records the exact typing duration.
- **Leaderboard Module:** Ensured that user scores are updated correctly in the MySQL database and displayed properly on the GUI.
- **Quiz Module:** Tested for correct question retrieval, response evaluation, and score updating.

Tools Used: Manual testing, Python's unit test (if implemented).

2. Integration Testing

After successful unit testing, integration testing was performed to ensure that the different modules work together smoothly.

This testing verified the interaction between the frontend interface, application logic, and database layer.

Key Focus Areas:

- Data flow between GUI and backend database.
- Proper fetching and storing of user scores and leaderboard updates.
- Smooth transition between the typing test, quiz module, and results screen.

Result: All integrated components communicated successfully without data mismatch or system crashes.

3. System Testing

System testing was conducted on the fully integrated system to validate that all functional and non-functional requirements were met. It was carried out in an environment similar to the expected deployment setting.

Test Cases Included:

- Starting and completing a typing test.
- Handling of invalid or incomplete input.
- Accurate calculation and display of typing results.
- Database updates after each test.
- Accessibility of leaderboard and quiz functions.

Outcome: The system met all specified requirements and functioned correctly under normal and boundary conditions.

4. User Interface (UI) Testing

The interface was tested for usability, layout, readability, and responsiveness. The goal was to ensure that users could easily interact with the system without confusion or difficulty.

Tests Performed:

- Button alignment and functionality.
- Text visibility and font consistency.
- Error and success message clarity.
- Responsiveness of GUI components.

Feedback: Users found the interface simple, intuitive, and visually appealing.

5. Performance Testing

Performance testing was done to evaluate how the system behaves under varying loads, such as multiple users accessing the leaderboard simultaneously or rapid repeated testing.

Aspects Tested:

- System response time.
- Database query efficiency.
- Resource utilization (CPU/memory).

Result: The application performed efficiently under normal usage conditions, with minimal lag and quick data retrieval.

6. Security Testing

Security testing was performed to ensure that the database and user data were protected against unauthorized access.

Key Points Tested:

- Restricted access to the MySQL database.
- Validation of user inputs to prevent SQL injection or crashes.
- Secure handling of user information and test results.

Result: The system maintained proper data security and input validation.

Test Case ID	Test Description	Input	Expected Output	Result
TC01	Start typing test	Click "Start"	Timer begins, text appears	Passed
TC02	Calculate typing speed	Type full passage	Correct WPM displayed	Passed
TC03			Accuracy % shown Calculate accuracy	Include typos Passed correctly
TC04	Store results	Submit test	Record saved in database	Passed
TC05	Display leaderboard	View leaderboard	Sorted results shown	Passed
TC06	Access quiz module	Select "Quiz Mode"	Quiz questions displayed	Passed
TC07	Invalid input handling	Leave input blank	Error message shown	Passed
TC08	Database connection	Connect to MySQL	Connection established	Passed

SECURITY MECHANISM

Security is important to ensure that user data, scores, and personal information are protected from unauthorized access or tampering. The following security mechanisms can be implemented in the Typing Speed Tester project:

1. User Authentication

- Users can create accounts with a username and password.
- Only registered users can submit scores or participate in quiz competitions.
- Passwords should be stored in an encrypted or hashed form (e.g., using hashlib in Python) to prevent unauthorized access.

2. Data Validation

- Input fields (username, typed text) are validated to prevent incorrect or malicious data entry.
- Prevents system crashes due to unexpected inputs.

3. Secure Data Storage

- User scores, leaderboard data, and history are stored in files or databases securely.
- Use file permissions to restrict access to sensitive files.
- Future versions can implement database security measures (e.g., SQLite with proper access control).

4. Error Handling and Exception Management

- Proper exception handling ensures that errors (like missing files or invalid inputs) do not expose sensitive data or crash the system.

5. Session Security (if GUI or Web-based in future)

- Ensure that user sessions expire after inactivity to prevent unauthorized access.
- Avoid storing plain text passwords or sensitive data in memory longer than needed.

6. Backup and Data Integrity

- Maintain regular backups of leaderboard and user history to prevent data loss.
- Verify data integrity when reading and writing files to prevent corruption.

COST ESTIMATION

Cost estimation helps to determine the total effort, resources and budget required for developing Typing Speed Tester software. It ensures proper planning and allocation of time, manpower and tools.

For this project, the COCOMO (Constructive Cost Model) is used to estimate effort, development time and cost.

Cost Estimation Model Used – COCOMO Model

COCOMO is an empirical software estimation model that calculates:

- Effort (E) – Person-months
- Development Time (D) – Months
- People Required (P)
- Total Cost

Since this project is small and academic, it falls under the Basic COCOMO – Organic Model Category.

COCOMO Formulas

- Effort (E) = $2.4 * (KLOC)^{1.05}$
- Development Time (D) = $2.5 * (E)^{0.38}$
- Personnel (P) = E/D

Size of Project

- LOC = 335
- KLOC = $335/1000 = 0.335$ KLOC

Calculations

Step 1 – Effect (E)

$$E = 2.4 * (0.335)^{1.05}$$

$$(0.335)^{1.05} \sim 0.317$$

$$E = 2.4 * 0.317$$

$$E \sim 0.76 \text{ Person-Months}$$

Step 2 – Development Time (D)

$$D = 2.5 * (0.75)^{0.38}$$

$$(0.75)^{0.38} \sim 0.88$$

$$D \sim 2.5 * 0.88$$

$$D \sim 2.2 \text{ Months}$$

Step 3 – Personnel Required (P)

$$P = E/D$$

$$P = 0.34 \text{ Developer}$$

Cost Calculation

Assume student developer cost = Rs 15,000 per month

Total cost = Effort * Cost per Month

$$= 0.76 * 15,000$$

$$= \text{Rs } 11,400$$

FUTURE SCOPE

The Typing Speed Tester project provides a foundation for improving typing skills, tracking performance, and encouraging healthy competition. In the future, the project can be enhanced with the following features and improvements:

1. Graphical User Interface (GUI) Enhancement

- Transition from a console-based interface to a user-friendly GUI using Tkinter, PyQt, Enhancement
- Include interactive buttons, colorful progress bars, and visual feedback for errors and performance.

2. Customizable Typing Tests

- Allow users to select difficulty levels, test durations, and themes (e.g., coding text, general knowledge, quotes).
- Include adaptive tests that adjust difficulty based on user performance.

3. Mobile or Web Version

- Develop the application as a web app or mobile app to make it accessible on multiple platforms.
- Enable cloud-based storage for user data and leaderboards.

4. Integration of Gamification Elements

- Introduce badges, achievements, and levels to motivate users.
- Add time-based challenges, streaks, and rewards for continuous improvement.

5. Support for Multiple Languages

- Allow typing tests in different languages to cater to a wider user base.

- Include bilingual or multilingual typing practice options.

6. Voice Typing and AI Assistance (Advanced)

- Incorporate voice-to-text typing for training speed and accuracy.
- Use AI to suggest tips for improving typing style and reducing errors.

APPENDICES

Appendix – A: Source Code of Application

This appendix contains the complete Python source code for the Typing Speed Tester, Competition Mode, Quiz Competition and Leaderboard System.

This application is built using:

- Python
- Tkinter for GUI
- SQLite3 for database
- Matplotlib for charts

Appendix – B: Modules and Libraries used

1. Tkinter

Used for building the entire GUI.

Function Used:

- Tk, Label, Button, Text, Frame, Radiobutton, Toplevel
- MessageBox, simpledialog

2. Sqlite3

Used to create and store leaderboard data in a local database.

Key functions:

- Connect ()
- Cursor ()
- Execute ()
- Commit ()

3. Time

The time module is used in the project for timer control, performance measurement, and time tracking.

Function Used:

Time.time () Used

to:

- Store the start time when typing begins.□
- Calculate total elapsed time by subtracting start time from current time.□
- Elapsed time is used to calculate WPM using:□
- $$WPM = \frac{\text{Words Typed}}{\text{Time}}$$
□
- Timer starts only when the user types the first key, preventing unfair counting.□

4. random

The random module is used in the application to introduce unpredictability and fairness in both typing tests and quizzes.

- i. random.choice ()
 - Selects a random typing paragraph each time the user starts or moves to the next paragraph.
 - Ensures users don't memorize paragraphs, maintaining fairness.
- ii. random.sample ()
 - Used for quiz competition.
 - Randomizes the order of questions, so player cannot predict the next questions.

5. Matplotlib

It is used for generating visual performance charts in the leaderboard window. It adds powerful data visualization component to the project.

Create line graphs for:

- WPM Scores
- Accuracy percentage

6. FigureCanvasTkAgg

- It is used to embed a matplotlib chart directly inside Tkinter window.
- Allows matplotlib figures to be displayed inside GUI, instead of a separate pop-up.
- Makes the leaderboard window visually rich and integrated.

Appendix – C: Database Schema (SQLite)

Database Name: leaderboard.db

Table: leaderboard

Column Name	Data Type	Description
name	TEXT	Player's name
wpm	INTEGER	Words per minute score
accuracy	REAL	Accuracy percentage

SQL Query Used:

```
CREATE TABLE IF NOT EXISTS leaderboard
```

```
(  
    name TEXT, wpm INTEGER,  
    accuracy REAL  
);
```

Appendix – D: Sample Test Cases

Test Case 1 – Calculate WPM

Input	Expected Output
Typed 45 words in 30 seconds	WPM = 90

Test Case 2 – Accuracy Calculation

Original Text	Typed Text	Expected Output
“Hello”	“Helli”	80% accuracy

Test Case 3 – Insert Into Leaderboard

Input	Expected Result
Name =Aditi, WPM = 72 Accuracy = 90%	Record added successfully.

Test Case 4 – Competition Mode

Input	Expected Result
3 players	Application takes each player turn one by one

Appendix – E: Functional Description of Major Functions

- **Handle_typing ()** ○ Detect user keystrokes ○ Starts timer automatically ○ Highlights correct/incorrect characters in real-time.
- **Check_result ()** ○ Stops timer ○ Calculates WPM ○ Handles competition and normal mode
- **Calculate_accuracy ()** ○ Compares original text with typed text and returns accuracy percentage.
- **Start_competition ()** ○ Takes number of players ○ Collect names ○ Start multi-typing challenge
- **Show_leaderboard ()** ○ Fetches top 10 scores

Display them in GUI

Shows performance chart using matplotlib

-

- **Start_quiz_competition ()**

Runs quiz for 1 – 5 players

- Displays score and winner

CODE

```
import tkinter as tk from tkinter import messagebox, simpledialog
import time, random, sqlite3 import matplotlib.pyplot as plt from
matplotlib.backends.backend_tkagg import FigureCanvasTkAgg

# ----- DATABASE SETUP -----
conn = sqlite3.connect('leaderboard.db') cursor =
conn.cursor() cursor.execute("""
CREATE TABLE IF NOT EXISTS leaderboard (
name TEXT, wpm INTEGER, accuracy REAL
)
""")
conn.commit()

# ----- PARAGRAPHS FOR TYPING ----- paragraphs =
[
("Technology has become an integral part of modern life. From smartphones to artificial "
"intelligence, our daily routines are increasingly driven by digital tools. The more we "
"understand and use technology responsibly, the more we can harness its potential to improve "
"education, healthcare, and communication across the globe."),
("Discipline is the bridge between goals and achievement. It requires commitment, focus, and "
"the ability to stay consistent even when motivation fades. Whether you are learning to code, "
"playing a musical instrument, or improving your health, discipline ensures long-term success."),
("Artificial intelligence continues to evolve rapidly, impacting industries like healthcare, "
"finance, and entertainment. Machine learning algorithms analyze massive datasets to make "
"accurate predictions, automate tasks, and personalize user experiences. The future will likely "
"see even greater collaboration between humans and intelligent systems."),
("Reading books not only improves your knowledge but also enhances your imagination and empathy. "
```

"When you read, you explore different perspectives and understand diverse cultures. In an era "

"dominated by short attention spans, dedicating time to read can truly enrich your mind."),

("The environment we live in is delicate and requires protection. Every small effort, such as "

"reducing waste, conserving water, and planting trees, contributes to a healthier planet. "

"Future generations depend on the actions we take today to preserve the beauty of nature."),

("The key to success in any field lies in continuous learning. The world is constantly changing, "

"and those who adapt will thrive. By staying curious and open-minded, you build the flexibility "

"needed to face challenges and seize new opportunities.")

]

----- QUIZ QUESTIONS ----- quiz_questions =

[

{ "question": "What does CPU stand for?",

"options": ["Central Process Unit", "Central Processing Unit", "Computer Personal Unit", "Central Processor Utility"],

"answer": "Central Processing Unit"},

{ "question": "Which programming language is known as the backbone of web development?",

"options": ["Python", "C++", "JavaScript", "Ruby"],

"answer": "JavaScript"},

{ "question": "What does GUI stand for?",

"options": ["Graphical User Interface", "General User Input", "Global Utility Index", "Graphical Unit Integration"],

"answer": "Graphical User Interface"},

{ "question": "Which company developed Python?",

"options": ["Google", "Microsoft", "CWI", "IBM"],

"answer": "CWI"},

{ "question": "What symbol is used to comment in Python?",

"options": ["//", "#", "<! >", "/* */"],

"answer": "#"}

]

----- MAIN APP CLASS -----

```
class TypingSpeedTester: def _init_(self, root):
    self.root = root
```

```

self.root.title("Typing Speed Tester + Competition + Quiz")

self.root.geometry("950x750")

self.root.config(bg="#1e1e2f")    self.start_time    =    0

self.elapsed_time    =    0    self.timer_running    =    False

self.sample_text = random.choice(paragraphs) self.players =

[] self.current_player_index = 0 self.results = [] self.typed_text

= "" self.create_widgets()

# ----- UI SETUP ----- def

create_widgets(self):

tk.Label(self.root, text="Typing Speed Tester", font=("Arial", 28, "bold"), fg="#fff",
bg="#1e1e2f").pack(pady=10) self.timer_label = tk.Label(self.root, text="Time: 0.00s",
font=("Arial", 16, "bold"), fg="cyan", bg="#1e1e2f") self.timer_label.pack(pady=5) frame =
tk.Frame(self.root, bg="#1e1e2f") frame.pack(pady=10) self.text_widget = tk.Text(frame,
height=10, width=95, font=("Consolas", 14), wrap="word", bg="#2b2b3c", fg="white",
insertbackground="white") self.text_widget.pack() self.text_widget.tag_configure("correct",
foreground="white") self.text_widget.tag_configure("incorrect", foreground="red")
self.text_widget.tag_configure("sample", foreground="#666") self.text_widget.insert("1.0",
self.sample_text) self.text_widget.config(state="disabled") self.root.bind("<Key>",
self.handle_typing) btn_frame = tk.Frame(self.root, bg="#1e1e2f") btn_frame.pack(pady=20)

```

```

tk.Button(btn_frame, text="Submit", command=self.check_result, bg="#4CAF50",
fg="white", font=("Arial", 14)).grid(row=0, column=0, padx=5) tk.Button(btn_frame,
text="Next Paragraph", command=self.next_sentence, bg="#9C27B0", fg="white",
font=("Arial", 14)).grid(row=0, column=1, padx=5) tk.Button(btn_frame,
text="Leaderboard", command=self.show_leaderboard, bg="#008CBA", fg="white",
font=("Arial", 14)).grid(row=0, column=2, padx=5) tk.Button(btn_frame, text="Typing
Competition", command=self.start_competition, bg="#FF9800", fg="white",
font=("Arial", 14)).grid(row=0, column=3, padx=5) tk.Button(btn_frame, text="Quiz
Competition", command=self.start_quiz_competition, bg="#FF5722", fg="white",
font=("Arial", 14)).grid(row=0, column=4, padx=5)

```

```

# ----- TIMER -----

```

```

def update_timer(self):
    if self.timer_running:
        self.elapsed_time = time.time() - self.start_time
        self.timer_label.config(text=f"Time: {self.elapsed_time:.2f}s")
        self.root.after(100, self.update_timer)

```

```

# ----- TYPING LOGIC -----

```

```

def handle_typing(self, event):
    if len(event.char) == 0:
        return
    if not self.timer_running:
        self.start_time = time.time()
        self.timer_running = True
        self.update_timer()
    if event.keysym == "BackSpace":
        self.typed_text = self.typed_text[:-1]
    else:
        self.typed_text += event.char
        self.text_widget.config(state="normal")
        self.text_widget.delete("1.0", tk.END)
        for i, char in enumerate(self.sample_text):
            if i < len(self.typed_text):

```



```

if self.typed_text[i] == char:
self.text_widget.insert(tk.END, char, "correct") else:
self.text_widget.insert(tk.END, char, "incorrect") else:
self.text_widget.insert(tk.END, char, "sample") self.text_widget.config(state="disabled")

# ----- CALCULATE RESULT -----
def check_result(self): if
not self.typed_text:
messagebox.showwarning("Empty", "Please type something before submitting!")
return self.timer_running = False total_time = time.time() - self.start_time words =
self.typed_text.split() wpm = round(len(words) / (total_time / 60)) accuracy =
self.calculate_accuracy(self.sample_text, self.typed_text)
# Competition mode active
if self.players:
current_player = self.players[self.current_player_index] self.results.append((current_player, wpm,
accuracy)) messagebox.showinfo("Result", f"{current_player}'s Score:\nWPM: {wpm}\nAccuracy:
{accuracy}%") self.current_player_index += 1 if self.current_player_index < len(self.players):
self.next_turn() else:
self.show_competition_results()
return
# Normal mode name = simpledialog.askstring("Name", "Enter your name for
leaderboard:") if name:
cursor.execute("INSERT INTO leaderboard VALUES (?, ?, ?)", (name, wpm, accuracy))

```

```

conn.commit() messagebox.showinfo("Result", f"Time: {round(total_time, 2)}s\nWPM: {wpm}\nAccuracy:
{accuracy}%") self.next_sentence() def calculate_accuracy(self, original, typed): correct = sum(1 for o, t in
zip(original, typed) if o == t) return round((correct / len(original)) * 100, 2) def next_sentence(self):
self.sample_text = random.choice(paragraphs)
self.typed_text = "" self.timer_running = False
self.elapsed_time = 0
self.timer_label.config(text="Time: 0.00s")
self.text_widget.config(state="normal")
self.text_widget.delete("1.0", tk.END)
self.text_widget.insert("1.0", self.sample_text,
"sample") self.text_widget.config(state="disabled")

# ----- TYPING COMPETITION -----

def start_competition(self):
num_players = simpledialog.askinteger("Competition", "Enter number of players (2-5):", minvalue=2,
maxvalue=5) if not num_players:
return self.players = []
self.results = []
self.current_player_index = 0

for i in range(num_players):
name = simpledialog.askstring("Player Name", f"Enter name for Player {i + 1}:") if
name:
self.players.append(name) messagebox.showinfo("Competition", "Each player will type the same
paragraph.\nLet's begin!") self.sample_text = random.choice(paragraphs) self.next_turn() def
next_turn(self):

```

```

player = self.players[self.current_player_index] self.typed_text = "" self.timer_running =
False self.elapsed_time = 0 self.timer_label.config(text="Time: 0.00s")
self.text_widget.config(state="normal") self.text_widget.delete("1.0", tk.END)
self.text_widget.insert("1.0", self.sample_text, "sample")
self.text_widget.config(state="disabled") messagebox.showinfo("Next Turn", f" {player},
it's your turn! Click OK to start typing.") def show_competition_results(self):
results_text = "\n".join([f" {name}: {wpm} WPM ( {acc}% accuracy)" for name, wpm, acc in self.results])
winner = max(self.results, key=lambda x: (x[1], x[2])) messagebox.showinfo("Competition Over", f" Results
\n\n {results_text} \n\n Winner: {winner[0]} ") for name, wpm, acc in self.results:
cursor.execute("INSERT INTO leaderboard VALUES (?, ?, ?)", (name, wpm, acc))
conn.commit() self.players = [] self.results = [] self.current_player_index = 0
self.next_sentence()
# ----- QUIZ COMPETITION ----- def
start_quiz_competition(self):
num_players = simpledialog.askinteger("Quiz Competition", "Enter number of players (1-5):", minvalue=1,
maxvalue=5) if not num_players:
return players = [] for i in
range(num_players):
name = simpledialog.askstring("Player Name", f"Enter name for Player {i+1}:") if
name:
players.append({"name": name, "score": 0}) for
player in players:

```

```

for q in random.sample(quiz_questions, len(quiz_questions)):

q_win = tk.Toplevel(self.root) q_win.title(f'Quiz —
{player['name']}') q_win.geometry("500x400")

q_win.config(bg="#1e1e2f") tk.Label(q_win,
text=q["question"], fg="white", bg="#1e1e2f",
wraplength=480, font=("Arial", 14, "bold")).pack(pady=20)

answer_var = tk.StringVar(value="") for option in

q["options"]:

tk.Radiobutton(q_win, text=option, variable=answer_var, value=option,
bg="#2b2b3c", fg="white", selectcolor="#1e1e2f", font=("Arial",
12)).pack(anchor="w", padx=40, pady=5) def submit_answer(): if
answer_var.get() == q["answer"]:

player["score"] += 1 q_win.destroy() tk.Button(q_win, text="Submit",
command=submit_answer, bg="#4CAF50", fg="white", font=("Arial",
12)).pack(pady=15) q_win.wait_window() results = sorted(players,
key=lambda x: x["score"], reverse=True) result_text =
"\n".join([f'{p['name']}: {p['score']} points' for p in results]) winner =
results[0]["name"]

messagebox.showinfo("Quiz Results", f' Quiz Competition Results \n\n{result_text}\n\nWinner: {winner}
")

# ----- LEADERBOARD ----- def

show_leaderboard(self):

leaderboard_win = tk.Toplevel(self.root) leaderboard_win.title("Leaderboard")
leaderboard_win.geometry("750x550") leaderboard_win.config(bg="#222") tk.Label(leaderboard_win,
text=" Leaderboard ", font=("Arial", 20, "bold"), fg="gold", bg="#222").pack(pady=10)
cursor.execute("SELECT * FROM leaderboard ORDER BY wpm DESC, accuracy DESC LIMIT 10")

```

```

records = cursor.fetchall() if not records: tk.Label(leaderboard_win, text="No records yet.", fg="white", bg="#222",
font=("Arial", 14)).pack(pady=10) return for i, (name, wpm, acc) in enumerate(records, start=1):
tk.Label(leaderboard_win, text=f"{i}. {name} — {wpm} WPM ({acc}% accuracy)", fg="white", bg="#222",
font=("Arial", 12)).pack(anchor="w", padx=20, pady=2) names = [r[0] for r in records] wpm_scores = [r[1] for r
in records] acc_scores = [r[2] for r in records] fig, ax = plt.subplots(figsize=(7, 3.5), facecolor="#222")
ax.plot(names, wpm_scores, color="#00FFFF", marker='o', linewidth=2, markersize=8, label="WPM")
ax.plot(names, acc_scores, color="#FF6347", marker='s', linewidth=2, markersize=8, label="Accuracy (%)")
top_index = wpm_scores.index(max(wpm_scores)) ax.plot(names[top_index], wpm_scores[top_index],
marker='o', markersize=12, color="#FFD700") ax.set_facecolor("#2b2b3c") ax.grid(True, linestyle='--',
alpha=0.5) ax.set_title("Top Players Performance", color="white", fontsize=16) ax.tick_params(axis='x',
colors='white') ax.tick_params(axis='y', colors='white') ax.legend(facecolor="#333", edgecolor="white",
fontsize=10) canvas = FigureCanvasTkAgg(fig, master=leaderboard_win) canvas.draw()
canvas.get_tk_widget().pack(pady=10) # ----- RUN APP -----

if __name__ == "__main__":
root = tk.Tk() app =
TypingSpeedTester(root)
root.mainloop()

```

OUTPUT

The screen which comes as an output when we run the above code.



Timer will start when we press first letter of the paragraph appearing on the screen.



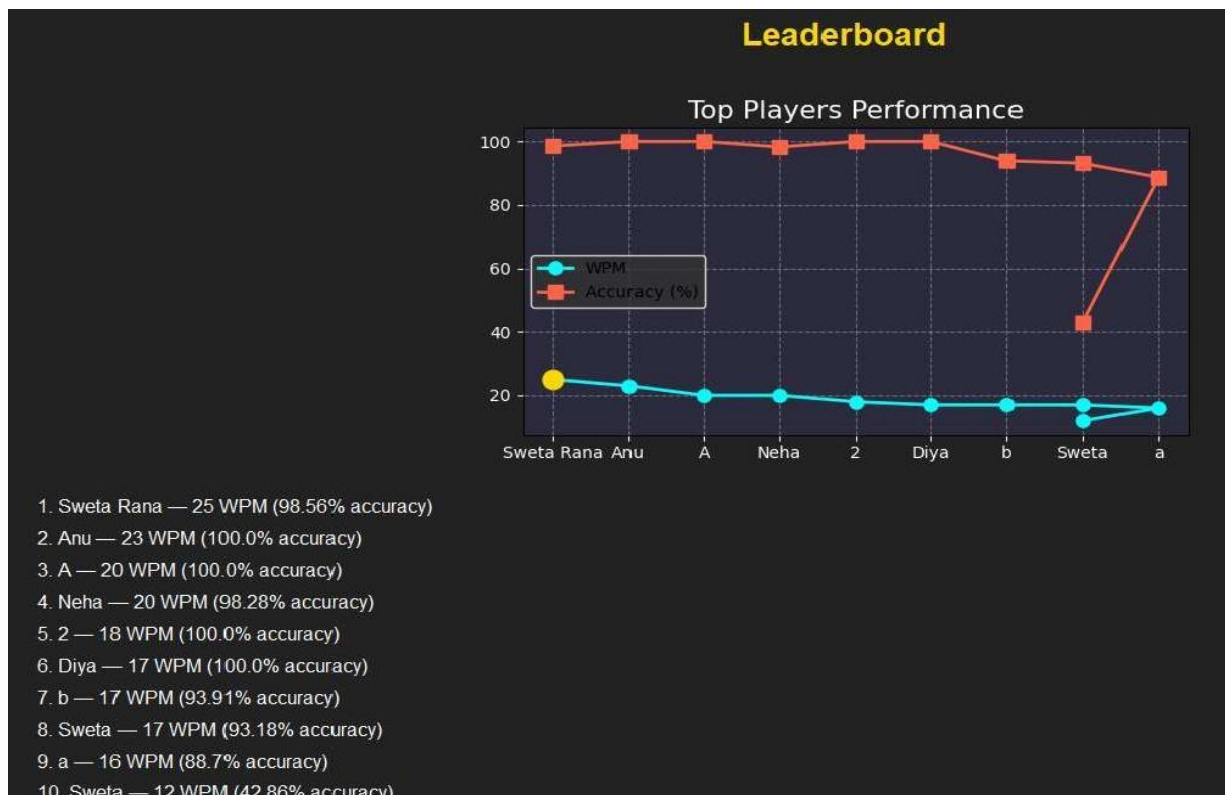
When we type wrong words it will show in red colour.

Typing Speed Tester

Time: 53.16s

Technology has become an integral part of modern life. From smartphones to artificial intelligence, our daily routines are increasingly driven by digital tools. The more we understand and use technology responsibly, the more we can harness its potential to improve education, healthcare, and communication across the globe.

The graph will show the top 10 best scores.



When we press typing competition first, this screen will be appear in which we have to choose the number of participant.

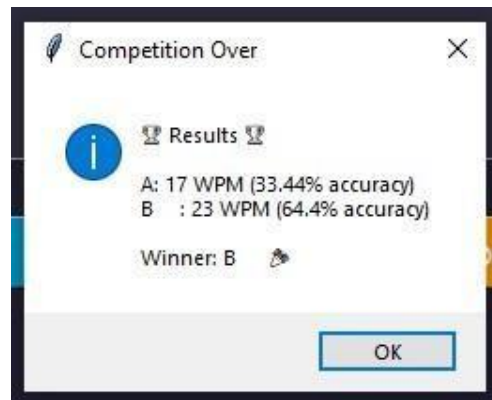


After that we have to write name of the players.

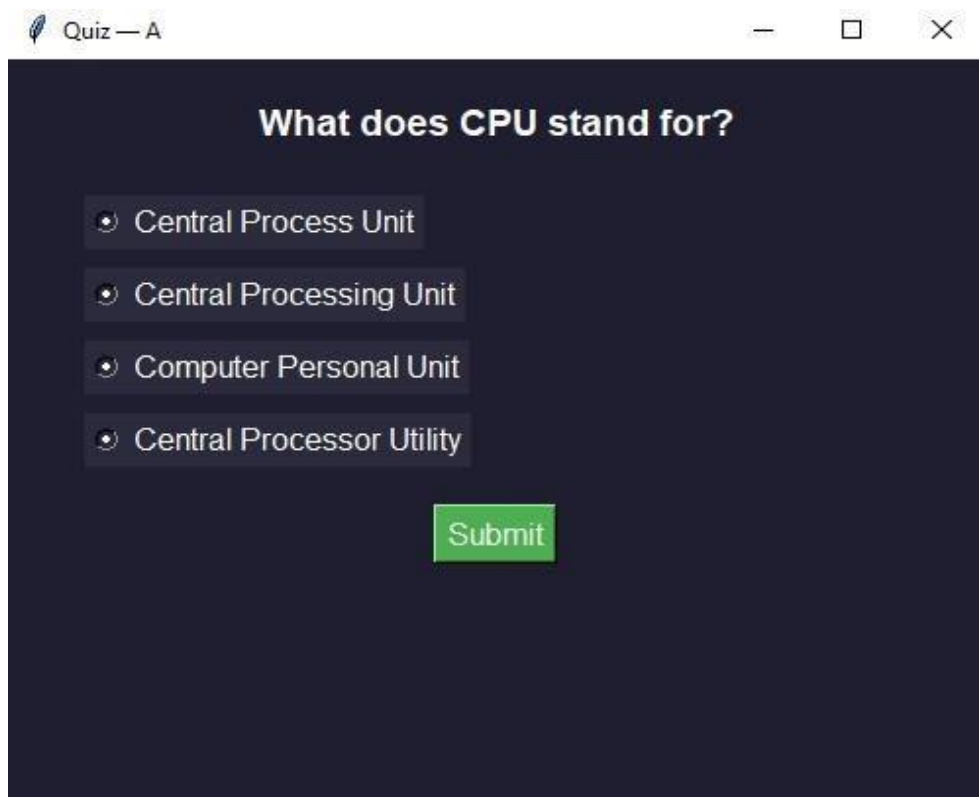


After that both the player has to type the same paragraph and it will show result as:

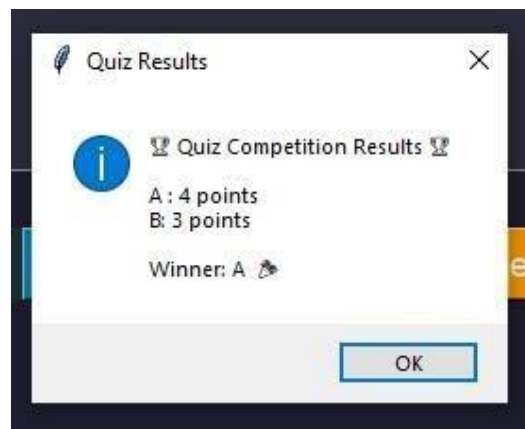




When we click on the quiz competition key then we have to choose number of players just like typing competition mode.



It show result as,



Leaderboard Table: leaderboard.db

	name	wpm	accuracy
	Filter	Filter	Filter
1	Sweta	12	42.86
2	2	18	100.0
3	vaibhavi	1	85.71
4	Sonam	11	62.5
5	Sweta	17	93.18
6	Diya	17	100.0
7	A	20	100.0
8	Neha	20	98.28
9	Anu	23	100.0
10	a	16	88.7
11	b	17	93.91
12	Sweta Rana	25	98.56
13	stuti	9	12.17
14	A	17	33.44
15	B	23	64.4

BIBLIOGRAPHY

1. Python Software Foundation: Python Documentation.

Available at: <https://docs.python.org>

2. Tkinter Module – Official Documentation: Python Standard Library

Available at: <https://docs.python.org/3/library/tkinter.html>

3. Matplotlib Developers: Matplotlib Documentation.

Available at: <https://matplotlib.org/stable/documentation>

4. GeeksforGeeks (Python and Tkinter tutorials)

Available at: <https://www.geeksforgeeks.org>

5. Programiz Python Tutorials

Available at: <https://www.programiz.com/python-programming>

6. Stack Overflow

Available at: <https://www.stackoverflow.com>

PROFILE OF TWO IT COMPANY

IT Company Profile 1: Tata Consultancy Services (TCS)

1. Company Name:

Tata Consultancy Services (TCS)

2. Founded:

1968

3. Founder:

J. R. D. Tata & F. C. Kohli

4. Headquarters:

Mumbai, Maharashtra, India

5. About the Company:

TCS is one of the world's largest IT services and consulting companies. It provides a wide range of technology solutions to global clients. The company is known for delivering reliable software products, digital transformation projects, and IT-enabled services.

6. Major Services Offered:

- IT Consulting
- Cloud Services
- Cybersecurity Solutions
- Business Process Outsourcing (BPO)
- Artificial Intelligence & Machine Learning Solutions
- Data Analytics

7. Global Presence:

TCS operates in 55+ countries and serves clients across various industries such as banking, manufacturing, healthcare, telecom, education, and retail.

8. Workforce:

More than 600,000 employees worldwide.

9. Achievements:

- Ranked among the world's most valuable IT service brands.
- Consistently listed in Fortune 500 companies.
- Received numerous awards for innovation, sustainability, and workplace culture.

IT Company Profile 2: Infosys

1. Company Name:

Infosys Limited

2. Founded:

1981

3. Founders:

Narayan Murthy, Nandan Nilekani, Kris Gopalakrishnan, S. D. Shibulal, K. Dinesh, Ashok Arora, N. S. Raghavan

4. Headquarters:

Bengaluru, Karnataka, India

5. About the Company:

Infosys is a leading multinational IT services and consulting company. It is known for its high-quality software solutions, IT infrastructure management, and strong emphasis on innovation and research.

6. Major Services Offered:

- Application Development
- Cloud Computing
- Cybersecurity and Risk Management
- Digital Transformation
- Engineering Services
- Business Consulting
- Automation & AI Tools
- Data Management and Analytics

7. Global Presence:

Infosys has offices in 50+ countries and serves clients from sectors such as finance, automotive, education, healthcare, telecom, and energy.

8. Workforce:

Around 335,000 employees across the globe.

9. Achievements:

- Infosys was the first Indian company listed on NASDAQ.
- Recognized as one of the most ethical companies globally.
- Strong focus on research through Infosys Labs.
- Awarded for sustainability, corporate governance, and digital excellence.

